

Steps in Computational Pipeline for Detection of Ecological Thresholds

Computational Pipeline Summary

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March 5, 2026

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Core Computation Target

Goal

Construct a **Zoobenthic Community Index (ZCI)** that quantifies benthic community condition along a sediment-contamination gradient in the SCDRS corridor, and detect **ecological thresholds** in such relationships.

Recall the expected visualizatoion of ZCI vs. Pollution Score

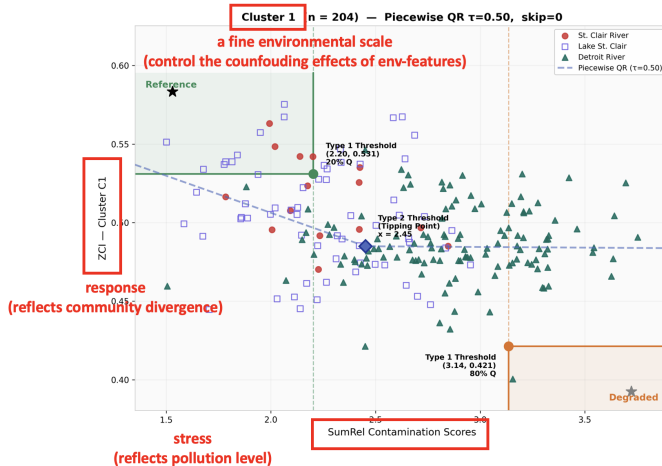


Figure: ZCI vs. Pollution Score (Stage 5 piecewise quantile regression, still under development)

The tone behind the computational stages

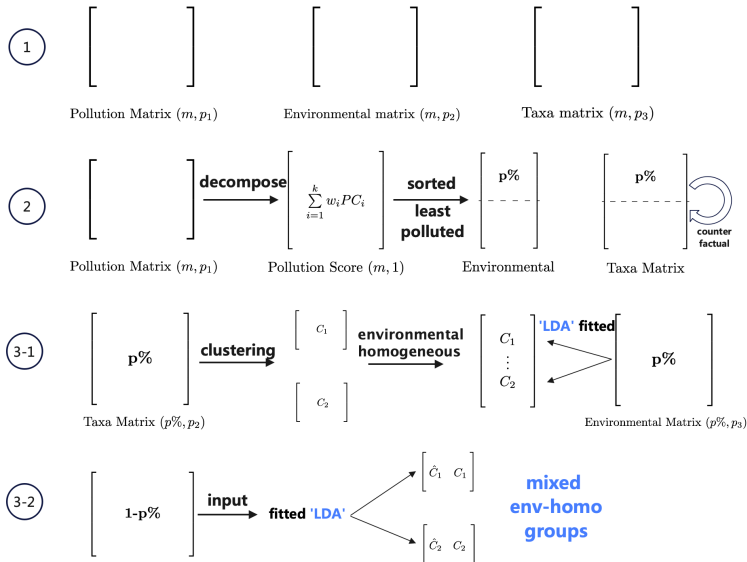
Goal

The computation work is very **open** now, we can update many steps and integrate extensions, meanwhile keeping the **integrity** of the current framework and results.

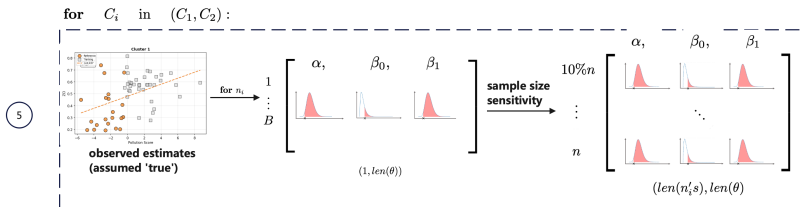
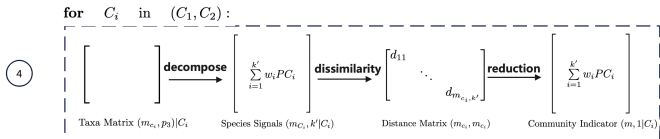
The framework to introduce each computation stage

- ➊ **Input data** — what specific data blocks and columns are used in this stage
- ➋ ***Specific operation** — transformation, model, normalization etc.
- ➌ **Outputs** — what results, artifacts are remained
- ➍ **Potential issues and improvements to apply**

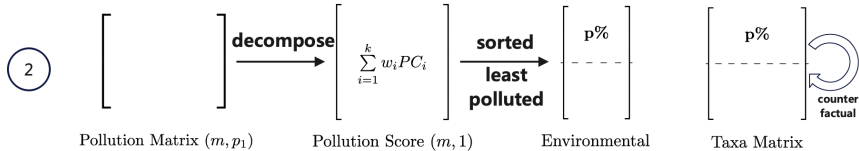
The summary of all computation stages



The summary of all computation stages



Stage 1: Pollution PCA — Overview



Stage 1: Pollution PCA — Input, Computation & Output

Input: Chemical matrix $C \in \mathbb{R}^{n \times 16}$ (16 sediment variables)

Computation:

- ① **Log₂ transform:** $\tilde{c}_{ij} = \log_2(1 + c_{ij})$
- ② **PCA** on $\tilde{C}$ (centred, *not* unit-scaled, $k=5$)
 - Scaled loadings: $L_{jm} = v_{jm} \sqrt{\lambda_m}$
 - Min-max scores: $s_{im}^* = \frac{s_{im} - \min}{\max - \min}$
- ③ **Composite score:** $s_i = \sum_{m=1}^k s_{im}^*$

Output: PC1–PC5, Pollution_Score

Stage 1: Design Notes & Potential Issues

Design Choice

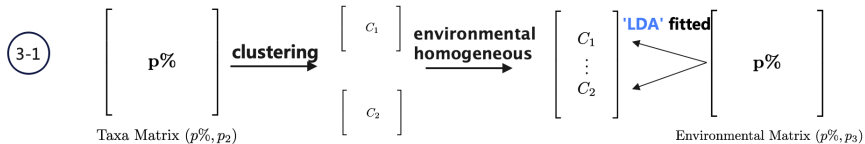
PCA on log-transformed data *without* unit-variance scaling $\Rightarrow$ high-variance contaminants dominate leading PCs.

Deliberate: preserves natural scale differences among pollutants.

Potential Issues

- No unit scaling — pollution score weighted toward high-variance contaminants; should be noted explicitly
- Equal-weight PC summation ($w_m=1$) — alternative weighting (e.g. by explained variance) not explored
- Min-max rescaling sensitive to outliers

Stage 2: Taxa Assemblage Clustering — Overview



Stage 2: Input, Computation & Output

Input:

- Taxa block $O \in \mathbb{R}^{n \times 16}$ (octave scale)
- Pollution Score s from Stage 1

Computation:

- 1 **Reference-site selection** — bottom 20% of pollution scores: $\mathcal{R} = \{i : s_i \leq s_{(\lfloor 0.20 n \rfloor)}\}$
- 2 **Taxa transform** — octave values used as-is (default)
- 3 **Ward's hierarchical clustering** on $O_{\mathcal{R}}$, Euclidean distance, $K=3$ initial clusters:

$$d_{\text{merge}}(A, B) = \sqrt{\frac{2|A||B|}{|A|+|B|}} \|\bar{o}_A - \bar{o}_B\|$$

Output: 02_updated_data.xlsx — Cluster labels for reference sites; non-reference sites remain NaN

Stage 2: Design Notes & Potential Issues

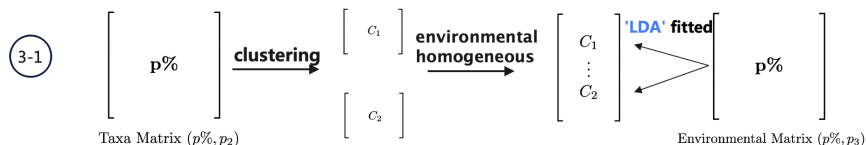
Design Choice

Ward's linkage minimises within-cluster variance — favours compact, spherical clusters in octave space. The 20% reference threshold is a *subjective* cut-off.

Potential Issues

- **Reference threshold sensitivity** — the 20% cut-off is arbitrary; varying it changes which sites are “reference”
- **Number of clusters K** — chosen by dendrogram inspection, not by a formal index (e.g. silhouette, gap statistic)
- **Octave-space distances** — Ward's uses Euclidean distance on octave values; alternatives such as Bray–Curtis are not applicable with Ward's linkage
- Small clusters may be artefacts ($\Rightarrow$ handled by hindsight merge)

Hindsight & ANOVA — Overview



Hindsight relabelling and ANOVA validation operate on Stage 2 outputs.

Hindsight: Input, Computation & Output

Input: Stage 2 cluster labels + environmental & taxa data (reference sites)

Computation:

- 1 **Cluster relabelling** — merge small clusters (e.g. $\{3 \rightarrow 1\}$) $\Rightarrow K=2$ habitat groups
- 2 **One-way ANOVA** per environmental variable & taxon:

$$F = \frac{SS_{\text{btw}}/(K-1)}{SS_{\text{wth}}/(n_{\text{ref}}-K)}$$

Optional pre-transforms: none | log | boxcox

Output: anova_env.xlsx, anova_taxa.xlsx, cluster_panel.png,
02_hindsight_updated_data.xlsx

Hindsight: Design Notes & Potential Issues

Role

ANOVA is *supportive* — it validates that habitat groups are ecologically distinct. It does **not** feed the ZCI computation.

Assumption Gap

- No **Shapiro–Wilk** test for within-group normality
- No **Levene / Brown–Forsythe** test for equal variances
- No Q–Q plots or residual diagnostics generated
- **Recommend:** add assumption checks; use Welch's ANOVA or Kruskal–Wallis H when assumptions fail; report effect sizes (η^2)

Stage 3: LDA Habitat Classification — Overview



Stage 3: Input, Computation & Output

Input:

- Environmental matrix $E_{\mathcal{R}}$ ($q=5$ variables: Depth, DO, Temperature, MPS, LOI)
- Cluster labels $g_{\mathcal{R}}$ from hindsight artifact
- Pollution Score s (for reference-site identification)

Computation:

- 1 **z-score standardisation** on reference sites: $e_{ij}^* = (e_{ij} - \bar{e}_j)/\sigma_j$
- 2 **LDA fitting** — maximise between/within-class variance: $LD_m = w_m^T e^*$
- 3 **Wilks' Λ** drop-one variable importance + Bartlett's χ^2 overall test
- 4 **Monte Carlo CV** — 1 000 stratified 80/20 splits
- 5 **Predict** all non-reference sites $\Rightarrow$ habitat labels

Output: 03_updated_data.xlsx — Predicted_Cluster, Is_Reference, LD coordinates, class probabilities

Stage 3: Design Notes & Potential Issues

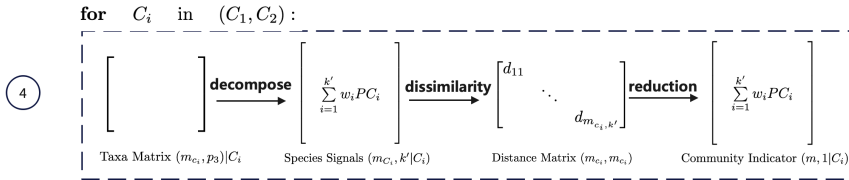
Design Choice

LDA assumes multivariate normality and equal covariance matrices. Wilks' Λ quantifies each variable's discriminatory contribution.

Potential Issues

- **CV data leakage** — StandardScaler fitted *globally before* CV splits; test folds have indirect knowledge of training means/variances $\Rightarrow$ CV accuracy slightly optimistic
- **Fix:** refit scaler inside each CV fold
- **Small sample size** — with $K=2$ and $|\mathcal{R}|$ reference sites, LDA stability depends on adequate per-class n
- **Multivariate normality** not formally tested

Stage 4: Bray–Curtis NMDS & ZCI — Overview



Stage 4: Input, Computation & Output

Input: Taxa O (octave), Pollution Score s, Predicted Cluster g

Computation (per cluster):

- 1 **Octave** $\rightarrow$ **relative abundance**: $\hat{p}_{ij} = \max(0, 2^{o_{ij}} - 0.0625)$, row-normalise
- 2 **Endpoint construction**: average the n_{ep} cleanest / most degraded sites $\Rightarrow$ synthetic reference & degraded centroids
- 3 **Bray–Curtis dissimilarity** (full pairwise matrix D): $d_{BC}(i, i') = \frac{\sum_j |p_{ij} - p_{i'j}|}{\sum_j (p_{ij} + p_{i'j})}$
- 4 **Iterative NMDS** — 3-pass SMACOF into $\mathbb{R}^2$; minimise stress
- 5 **PCA rotation + axis flip** (NMDS1 $\leftarrow -\text{NMDS1}$ if $r > 0$)
- 6 **Weighted-average species scores**: $WA_j = \sum_i p_{ij} x_i / \sum_i p_{ij}$
- 7 **ZCI (BC-Direct)**: $z_i = \frac{d_{BC}(i, \mathcal{E}_{deg})}{d_{BC}(i, \mathcal{E}_{ref}) + d_{BC}(i, \mathcal{E}_{deg})}$

Output: 04_updated_data.xlsx — NMDS1, NMDS2, ZCI, ZCI_Method

Stage 4: Where Bray–Curtis Is Applied (Twice)

Bray–Curtis enters at **two distinct points**:

1. For NMDS Ordination

Full pairwise D among all cluster sites + 2 endpoints.

D feeds the SMACOF algorithm $\Rightarrow$ 2-D embedding.

Purpose: *visualisation* of community structure.

2. For BC-Direct ZCI

Each site vs. endpoint *centroids* only.

Operates directly on P — **independent** of NMDS.

Purpose: the *actual* community condition index.

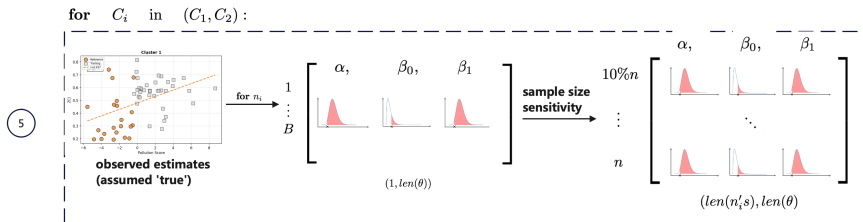
Cluster	n_{ep}	Method
1	15	BC-Direct
2	3	BC-Direct

Stage 4: Design Notes & Potential Issues

Issues in Current Implementation

- **NMDS axis flip is enforced, not discovered** — NMDS1 is flipped so $r(\text{NMDS1}, s) < 0$; the negative correlation is an *artefact of convention*, not evidence of a contamination gradient
- **Endpoint sensitivity** — small n_{ep} (e.g. 3 for Cluster 2) makes the centroid sensitive to individual outlier sites; ZCI stability under varying n_{ep} should be reported
- **NMDS stress** — if stress > 0.20 , the 2-D representation is unreliable; must be evaluated per cluster
- **Cluster-specific ZCI** — computed separately within each habitat cluster; cross-cluster ZCI comparisons require caution (different endpoint communities)
- **ZCI uses BC-Direct (not NMDS coords)** — more defensible but means NMDS serves only visualisation, not the index itself

Stage 5: Piecewise Quantile Regression — Overview



Stage 5: Input, Computation & Output

Input: Pollution Score s (S1), Predicted Cluster g (S3), ZCI z (S4)

Computation (per cluster, $\tau \in \{0.10, 0.15, \dots, 0.90\}$):

① **Piecewise linear quantile regression:**

$$Q_{\tau}(\text{ZCI} \mid \text{PS}=x) = \beta_0 + \beta_1 x + \beta_2 (x - \psi)_+$$

② **Profile grid search** over candidate breakpoints ψ ; LP sub-problem at each ψ_c

③ **Wild bootstrap CI** (Feng, He & Hu 2011) — Mammen's two-point weights, $B=500$ replicates, 90% CI

Output: 05_updated_data.xlsx — breakpoint $\hat{\psi}$, slope coefficients, CI bounds per τ

Note: Stage 5 is currently commented out in the corridor-wide runner; active in the Detroit River runner.

Stage 5: Design Notes & Potential Issues

Design Choice

Piecewise quantile regression detects *threshold breakpoints* in the ZCI–Pollution relationship at multiple quantile levels, capturing heterogeneous community responses.

Potential Issues

- **Grid resolution** for ψ — coarse grid may miss the optimal breakpoint; finer grid increases computation
- **Bootstrap B** — 500 replicates adequate for 90% CI but may be insufficient for 95% CI; consider $B \geq 1000$
- **Edge effects** — grid spans 10th–90th percentile; breakpoints near extremes are not detectable
- **Single breakpoint model** — only one hinge; multiple thresholds would require spline or multi-breakpoint extensions

Stage RDA: Redundancy Analysis (Complementary)

Purpose: Quantify how much taxa variation is explained by environmental predictors.

- 1 Multivariate OLS: $\hat{Y} = X_c \hat{B}$
- 2 PCA on fitted values $\Rightarrow$ constrained eigenvalues
- 3 Permutation tests (999 permutations):

Test	Null hypothesis
Global	No linear relationship between Y and X
Per-axis	Individual RDA axis non-significant
Per-term	Individual variable non-significant (Freedman-Lane)

Does not feed downstream stages — validation only.

Stage-by-Stage Artifact Chain

Stage	Input	New Columns	Output
S1 (PCA)	Original workbook	PC1–5, Pollution_Score	01_updated.xlsx
RDA	Orig + S1	<i>None (figs/tables)</i>	<i>No artifact</i>
S2 (Clust.)	Orig + S1	Cluster, Is_Reference	02_updated.xlsx
Hindsight	S2 + Orig	Relabelled Cluster	02_hind.xlsx
S3 (LDA)	Orig + S1 + Hind.	Pred. Cluster, LD1	03_updated.xlsx
S4 (NMDS)	Orig + S1 + S3	NMDS1/2, ZCI	04_updated.xlsx
S5 (PQR)	S1 + S3 + S4	Model params	05_updated.xlsx

TITAN: How It Complements the Current Framework

What TITAN Adds

A **taxon-resolution layer** the pipeline currently lacks: *which* taxa drive community shifts and at *what* pollution level.

Algorithm (per cluster):

- 1 Binary partitions along the pollution gradient
- 2 IndVal per taxon per split $\Rightarrow$ z-score vs. permutation null
- 3 Change-point: $x_c^{*(j)} = \arg \max_{x_c} z_j(x_c)$
- 4 Direction: z^- (sensitive) vs. z^+ (tolerant)
- 5 Bootstrap purity ≥ 0.95 & reliability ≥ 0.95
- 6 Community threshold: peak of $\text{sum}(z^\pm)$

A new method for detecting and interpreting biodiversity and ecological community thresholds

Matthew E. Baker^{1*} and Ryan S. King^{2*}

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Reference: Baker & King (2010)

TITAN: Integration Plan & Key Questions

Input from existing pipeline:

TITAN Input	Source	Stage
Gradient x	Pollution_Score	Stage 1
Taxa matrix Y	Relative abundance P	Octave inverse
Cluster grouping	Predicted_Cluster	Stage 3

Key questions TITAN answers:

- Does the 20% reference threshold match an **ecological change point**?
- Gradual taxon turnover or **abrupt regime shift**?
- Independent comparison with Stage 5 piecewise-QR breakpoint $\hat{\psi}$
- Which individual taxa are responsible for the community shift?

Execution: R package TITAN2 via rpy2; overlay $\text{sum}(z^{\pm})$ peaks as vertical lines on ZCI–Pollution plots

Summary of Recommendations

- 1 **ANOVA assumptions**: add Shapiro–Wilk + Levene’s tests; Welch/Kruskal–Wallis as robust alternatives
- 2 **LDA CV leakage**: move `StandardScaler` inside each fold
- 3 **NMDS axis-flip**: state explicitly that orientation is enforced
- 4 **Endpoint sensitivity**: report ZCI stability under varying n_{ep}
- 5 **PCA scaling**: note log-transformed but not unit-scaled
- 6 **TITAN**: embed per-cluster analysis after Stage 3; overlay thresholds on ZCI–Pollution plots

Questions?

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